Hedieh Abdollahi

hedieh.abdollahi@csfk.org — abdollahi.hedieh2013@gmail.com
ResearchGate — Google Scholar — Website — ADS

RESEARCH PROFILE

Astronomy researcher specialising in time-domain observations and computational methods to study resolved stellar populations and reconstruct their star-formation histories. My research focuses on long-period variables in the Local Group and RR Lyrae stars, combining time-series photometry, variable-star analysis, and stellar-evolution modelling. I use evolved stars to study stellar populations, star formation, mass loss, and dust production.

RESEARCH AFFILIATIONS

Affiliated Researcher Mar 2026 – Present
Konkoly Observatory, Budapest, Hungary

Visiting Research Fellow Mar 2026 – Present
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

POSTDOCTORAL POSITIONS

Postdoctoral Research Fellow 1 Nov 2024 – 28 Feb 2026
Konkoly Observatory, Budapest, Hungary

Postdoctoral Researcher 22 May 2023 – 30 Oct 2024
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

EDUCATION

PhD in Astronomy, Al-Zahra University, Tehran, Iran 2016 – 16 Mar 2023
Including an approved two-year break.

Master's Degree in Physics, Al-Zahra University, Tehran, Iran 2013 – 2015

Bachelor's Degree in Physics, Shahed University, Tehran, Iran 2007 – 2011

PUBLICATIONS

Peer-Reviewed Journal Articles

- Calculation of Dust Production by Long-Period Variable Stars in the Galaxy NGC 6822. *Iran. J. Phys. Res.*, 2026.
- A Study on the Evolution of NGC 6822. I. Investigating the Star Formation History through Evolved Stellar Populations. *ApJ*, 2026.
- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies VIII: A Census of Long-Period Variable Stars across the Andromeda Dwarf Satellite System. *ApJ*, 2026.
- Rare find: Discovery and chemo-dynamical properties of two s-process-enhanced RR Lyrae stars. *A&A*, 2025.
- Long-period variable stars in NGC 147 and NGC 185 – II. Their dust production. *ApJ*, 2025.
- Search for binary-channel metal-rich RR Lyrae candidates. *A&A*, 2025.
- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies VI: The Star Formation History and Dust Production in Andromeda IX. *ApJ*, 2023.

Manuscripts Under Review

- The UKIRT M33 Monitoring Project. VII. A 15-Year K-band Photometric Catalogue of Long-Period Variable Stars in the Central Kiloparsec of M33
- SEDust: A DUSTY-Based SED-Fitting Pipeline for Dust and Mass-Loss from AGB and LPV Stars

Manuscripts in Preparation

- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies IX: Star Formation History and Dust Production in Andromeda Dwarf Galaxies Traced by Long-Period Variable Stars
- Characterizing Time-Resolved RRab Light Curves to Reveal Additional Modes and Phase Variations
- Recovering the Star Formation History of M31 Using Long-Period Variable Stars

Conference Proceedings and Contributions

- The UKIRT M33 Monitoring Project: A 15-Year Near-Infrared Variable Star Catalogue for the Central Kiloparsec, and Prospects for Star Formation History and Dust Return, BAO
- A Deep Dive into Stellar Populations in M33's Central Region: Near-Infrared Observations and Analysis, BAO
- Half-Light Radius Measurements of Andromeda Dwarf Satellites from the Isaac Newton Telescope Survey Using Exponential, Plummer, and Sérsic Fits, ADS, BAO
- Star Formation History of the Local Group Dwarf Irregular Galaxy, NGC 6822, ADS
- Detection of the Long Period Variable Stars of And II Dwarf Satellite Galaxy, ADS
- Investigating the Dust Production in the Local Group Dwarf Irregular Galaxy, NGC 6822, Through Dusty Stellar Populations, ADS
- Investigating the Evolutionary History of the Local Group Dwarf Irregular Galaxy, NGC 6822, Through Long-Period Variable Stars, ADS
- The Isaac Newton Telescope Monitoring Survey of Local Group Dwarf Galaxies VIII: The Star Formation History and Dust Production Rate of Andromeda Dwarf Satellites, ADS
- Comprehensive Compilation of Stellar Data in M33's Central Region: Unveiling the Final-Stage Evolution Through Near-Infrared Survey, ADS
- Thorough Collection of Stellar Information in M33's Core: Revealing Late-Stage Evolution with Near-Infrared Study, ADS
- The Formation and Evolution of Andromeda IX, ADS
- The Role of LPVs in And IX, from Star Formation to Dust Production, ADS
- The Star Formation History and Dust Production in Andromeda IX, Proceedings of the International Astronomical Union, ADS, Cambridge Journal
- INT Monitoring Survey of Local Group Dwarf Galaxies: Star Formation History and Chemical Enrichment, ADS

Other Scholarly Contributions

- Physics Workbook for High School Students, January 2010 (Scientific Editor)

CONFERENCES, SCHOOLS, AND PRESENTATIONS

- Our Unstable Universe, Byurakan, Armenia, 14–18 September 2026 – *Contributed Talk*
- 5th Edition of the Why Galaxies Care About AGB Stars Conference Series: 3D Winds in the Cosmic Matter Cycle, Santiago, Chile, 17–21 November 2025 – *Contributed Talk*
- 403rd IAU Symposium, The Hidden Beauty of the Galactic Outskirts, Córdoba, Spain, 6–10 October 2025
- Astronomical Surveys and Big Data 3 (ASBD3), Byurakan, Armenia, 15–19 September 2025 – *Contributed Talk*
- I-HOW JWST Hands-On Workshop for Central and Eastern Europe, Toruń, Poland, 14–25 July 2025
- EAS Annual Meeting, Cork, Ireland, 23–27 June 2025 – *Poster*, Zenodo
- 6th Scientific Writing for Young Astronomers, Sintra, Portugal, 5–9 May 2025
- IAU South West and Central Asian Regional Office of Astronomy for Development (SWCA ROAD), Byurakan, Armenia, 16–20 September 2024 – *Contributed Talk*
- 9th Byurakan International Summer School for Young Astronomers (9BISS), Byurakan, Armenia, 9–13 September 2024
- EAS Annual Meeting, Padova, Italy, 1–5 July 2024 – *Poster*
- 22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun, San Diego, USA, 24–28 June 2024 – *Poster*
- 31st Physics Spring Conference (IPM), Tehran, Iran, 16 May 2024 – *Poster*
- What Was That? Planning ESO Follow-up of Transients, Variables, and Solar System Objects in the LSST Era, Garching, Germany, 22–26 January 2024 – *Poster*, Zenodo 1, Zenodo 2

- A Decade of ESO Wide-Field Imaging Surveys, Garching, Germany, 16–21 October 2023 – *Contributed Talk*, Video, Zenodo
- Stars (Across the Universe), Napoli, Italy, 16–20 October 2023 – *Poster*, Zenodo
- Asia-Pacific Regional IAU Meeting (APRIM), Koriyama, Japan, 7–11 August 2023 – *Poster*
- 376th IAU Symposium, Budapest, Hungary, 17–21 April 2023 – *Poster*
- International Spring School, Modern Methods of Cosmic Distance Determination, Konkoly Observatory, Budapest, Hungary, 12–15 April 2023
- XXXI General Assembly of the IAU (IAUGA 2022), IAUS 373, Busan, Republic of Korea, 2–11 August 2022 – *Contributed Talk*
- 5th INO Workshop on Transient Events and Multi-Messenger Astrophysics, Tehran, Iran, 28–29 July 2022
- 4th INO Workshop on Formation, Evolution, and Asteroseismology of Stars with a View to Exoplanets, Tehran, Iran, 19–20 May 2022
- Iranian National Observatory: 3.4-m Telescope Instrumentation and Science Cases, Tehran, Iran, 2–3 February 2022
- 3rd INO Workshop on Exoplanets and Their Identification, Tehran, Iran, 13–14 January 2022
- 2nd INO Workshop on Star Clusters and the Initial Mass Function, Tehran, Iran, 16–17 December 2021
- 1st INO Workshop on Formation, Evolution, and Structure of Galaxies, Tehran, Iran, 14–15 October 2021
- 2nd Regional Astronomical Summer School on Observational and Data Astronomy and Space Sciences, Byurakan, Armenia, 13–17 September 2021
- Workshop From Stars to Galaxies: Introduction to Modern Spectroscopic Methods, Al-Zahra University, Tehran, Iran, 4 January 2020
- Tehran Meeting on Cosmology (IPM), Tehran, Iran, 5–10 August 2017

SOFTWARE TOOLS

Photometry & image processing:	DAOPHOT/ALLSTAR, IRAF, Photutils, DS9, AstroImageJ
Time-series analysis:	Period04, astropy.timeseries (Lomb–Scargle), Lightkurve
SED modelling & radiative transfer:	DUSTY, DESK, *SEDust*
Astronomical survey tools:	TOPCAT, Aladin, Gaia Archive / TAP+, VizieR, IRSA, SkyMapper, MAST
Database query & automation:	astroquery, PyVO, SIMBAD
Version control & environment:	Git, GitHub, Conda, Docker
Operating systems:	Unix/Linux (Ubuntu, CentOS), macOS
Python scientific stack:	Python (NumPy, SciPy, Astropy, Matplotlib, Pandas, Seaborn, IPython, JupyterLab)

AWARDS AND SELECTED RESEARCH VISITS

- Short-term research visit to ESO, Santiago, Chile, 11–16 November 2025
- Dr. Samimi Prize for the Best PhD Thesis in Astrophysics and Cosmology in Iran, 2024
- Short-term research visit to Konkoly Observatory (HUN-REN CSFK), Budapest, Hungary, March–April 2024

SUPERVISION AND MENTORING

- **PhD Thesis Advising:** Unveiling Long-Period Variables in M33's Central Region: Insights into Stellar Evolution and Star Formation via Near-Infrared Photometry
- **Project Mentoring:** A Study on the Evolution of NGC 6822. I. Investigating the Star Formation History Through Evolved Stellar Populations

REFERENCES

- **Dr. Jacco Th. van Loon**
  Lennard-Jones Laboratories, Keele University, ST5 5BG, UK
  Email: j.t.van.loon@keele.ac.uk
- **Dr. László Molnár**

Konkoly Observatory, HUN-REN Research Centre for Astronomy and Earth Sciences, MTA Centre of Excellence, Konkoly-Thege Miklós út 15–17, H-1121, Budapest, Hungary
ELTE Eötvös Loránd University, Institute of Physics and Astronomy, 1117, Pázmány Péter sétány 1/A, Budapest, Hungary
Email: molnar.laszlo@csfk.org

- **Dr. Iain McDonald**
Department of Physical Sciences, The Open University, Walton Hall, Milton Keynes, UK
Jodrell Bank Centre for Astrophysics, Alan Turing Building, University of Manchester, M13 9PL, UK
Email: iain.mcdonald-2@manchester.ac.uk
- **Dr. Atefeh Javadi**
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
Email: atefehj@gmail.com